

A Content-Based Journal-Finder and Topic-Cluster Discovery System for Computer-Science Publications

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Abstract—Selecting an appropriate journal for a finished manuscript is a recurring problem for researchers. We address it as a content-based recommendation problem: given the abstract of a new computer-science article, return the five most relevant target journals from a corpus drawn from Web of Science. Working with the CompSciencePub database (23,034 articles after dropping empty abstracts and singleton journals, across 428 journals), we compare four standard recommenders over a TF-IDF representation of abstracts and keywords: a centroid cosine-similarity ranker, a k-nearest-neighbours article ranker, multinomial Naive Bayes, and one-vs-rest logistic regression. We evaluate them with Top-1, Top-3, Top-5 accuracy and Mean Reciprocal Rank on a stratified 80/20 hold-out. As a complementary unsupervised analysis, K-Means is applied to the same TF-IDF space, with the cluster count chosen by silhouette score. Logistic regression achieves Top-5 accuracy of 0.888, about 76 times the random baseline of $5/|J| \approx 0.012$. The clusters recover recognisable computer-science sub-areas (wireless sensor networks, telecommunications and electronics, hardware and VLSI, semantic web, library and information science, operations research, finite-element methods, software engineering and HCI, optimisation and evolutionary algorithms, theoretical computer science, cloud computing and security, computational biology, fuzzy decision making, and AI and neural learning). The system is packaged as a Python module plus a Jupyter notebook, exposing a single function `find_top_5_journals(abstract)`.

Keywords—journal recommendation, text classification, TF-IDF, topic modelling, K-Means, information retrieval, data mining

I. INTRODUCTION

The number of active scientific journals has grown rapidly, and researchers routinely struggle to identify the venue best matched to their manuscript's topic, scope and audience. A poor venue choice costs months of review time and suppresses the work's impact. Several commercial tools already exist (Elsevier's Journal Finder [10], Springer's Journal Suggester [11], Wiley's Journal Finder [12], JANE [6]), but they are closed-source and trained on private corpora.

This project re-implements the same idea on an open corpus of 23,034 computer-science articles published in 428 journals indexed by Web of Science, using only standard data-mining techniques covered in the course. The deliverable is a function that, given the abstract of a new paper, returns a ranked list of five candidate journals. As a secondary analysis we cluster the corpus to expose its dominant research themes; the cluster summaries provide a sanity check on the recommender and a useful overview of the dataset.

II. RELATED WORK

A. TF-IDF and the Vector Space Model

The TF-IDF weighting scheme of Salton and Buckley [1] remains the workhorse of text-classification baselines. Manning, Raghavan and Schütze's textbook [2] formalises the vector-space model and the cosine similarity that we use for the centroid recommender. TF-IDF retains a strong position on small and domain-specific corpora because it is interpretable, deterministic and orders of magnitude cheaper than dense neural embeddings.

B. Journal Recommenders

Elsevier's Journal Finder [10] matches a user-supplied title and abstract against an index of articles already published in Elsevier journals. Springer Nature's Journal Suggester [11] works similarly on Springer titles and Wiley offers an analogous service [12]. JANE [6] is an open academic tool that indexes PubMed abstracts. Wang et al. [7] report that a TF-IDF baseline is hard to beat for venue recommendation when the corpus is moderately sized; Kang et al. [8] extend this with citation-graph features. Our work follows the same content-only baseline strategy on a computer-science corpus.

C. Topic Modelling and Clustering

Two dominant traditions exist: hard-assignment clustering, of which K-Means [3] on TF-IDF vectors is the canonical example, and probabilistic mixture models such as Latent Dirichlet Allocation [4]. Selecting the number of clusters is itself a sub-problem; Rousseeuw's silhouette coefficient [5] is a standard internal-validation criterion which we adopt with cosine distance.

III. DATASET

The dataset is provided as the SQLite database CompSciencePub.sqlite. Its schema, summarised in the ER diagram of the assignment, has AcademicRecord as its central entity (one row per article) and links out to abstracts, publications (the journal label), author keywords, Web-of-Science Keywords Plus and subject categories.

The raw database contains 23,801 articles in 466 journals. After dropping articles with empty abstracts and journals with a single article (because stratified train/test splitting requires at least 2 samples per class), 23,034 articles across 428 journals remain. The class distribution is mildly imbalanced (Fig. 1): a large group of journals reach the Web-of-Science indexing cap of 76 articles, while a long tail of specialised journals contributes only a handful. Median abstract length is 159 words.

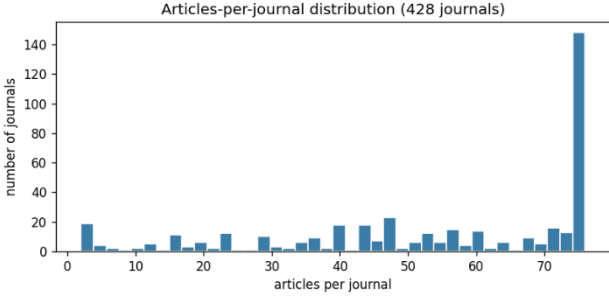


Fig. 1 Articles per journal in the working corpus. The peak at the right is the Web-of-Science indexing cap; a long tail of journals contributes only a few articles each.

For each article we build a single text field by concatenating the abstract, the author keywords, the Keywords-Plus tokens and the subject names, then apply a light cleaning step (lower-casing, keeping alphabetic tokens, removing English stop-words). The result is a single text column on which all downstream models operate.

IV. METHODOLOGY

A. Vector Representation

Each cleaned document is mapped to a TF-IDF vector with unigram and bigram features. We retain at most 15,000 features, drop terms appearing in fewer than two documents and apply sub-linear term-frequency scaling. The weight of term t in document d is given by $w(t,d) = (1 + \log \text{tf}(t,d)) \cdot \log((N+1) / (\text{df}(t)+1)) + 1$, followed by L2-normalisation of the document vector.

B. Recommenders

- **Centroid + Cosine.** For each journal j we compute the mean of its training-article vectors $c(j)$, L2-normalise it, and rank journals by the cosine similarity between the query abstract q and each centroid $c(j)$.
- **K-Nearest-Neighbours.** The query is compared against every training article. The 25 nearest articles cast similarity-weighted votes for their journals; the journal with the largest aggregate similarity is ranked first.
- **Multinomial Naive Bayes.** A classical generative text-classification baseline. Each journal is treated as a class and MNB is trained on the TF-IDF vectors. The five journals with the largest predicted probabilities are returned.
- **Logistic Regression.** A discriminative one-vs-rest L2-regularised logistic-regression model with the SAGA solver and $C = 4$.

C. Evaluation Protocol

We split the data 80/20 with stratification on the journal label. We report Top-1 accuracy (did we predict the ground-truth journal at rank 1?), Top-3 and Top-5 accuracy (did the ground-truth journal appear in the top 3 or 5 ranks?), and Mean Reciprocal Rank, $\text{MRR} = (1/N) \sum 1/\text{rank}(i)$. The deliverable demands Top-5 accuracy specifically, so that is the metric we use to pick the production model.

D. Topic Clustering

For the unsupervised view we re-vectorise the entire corpus and run K-Means with k in $\{4, 6, \dots, 24\}$. For each k we compute the silhouette score on a random subsample of

1,000 documents using cosine distance, and pick the k with the highest mean silhouette. We then describe each cluster by the 12 highest-weight terms in the K-Means centroid, and visualise the partition with a 2-component truncated-SVD projection.

V. EXPERIMENTS AND RESULTS

A. Top-k Accuracy

Table I summarises the four recommenders on the held-out 20% test set. All four substantially exceed the random baseline of $5/|J| \approx 0.012$ for Top-5, confirming that the abstract plus the keyword and subject metadata carry more than enough signal to narrow the journal choice from 428 candidates down to a handful. Logistic regression achieves the strongest result (Top-5 = 0.888); the centroid + cosine baseline is a close second (Top-5 = 0.855) at a small fraction of the training cost. Fig. 2 plots the same numbers as a grouped bar chart.

TABLE I
TEST-SET RANKING METRICS ON A STRATIFIED 80/20 HOLD-OUT

Model	Top-1	Top-3	Top-5	MRR
Centroid + Cosine	0.582	0.786	0.855	0.689
KNN (k=25)	0.425	0.607	0.675	0.522
Naive Bayes	0.437	0.612	0.683	0.530
Logistic Regression	0.639	0.827	0.888	0.737

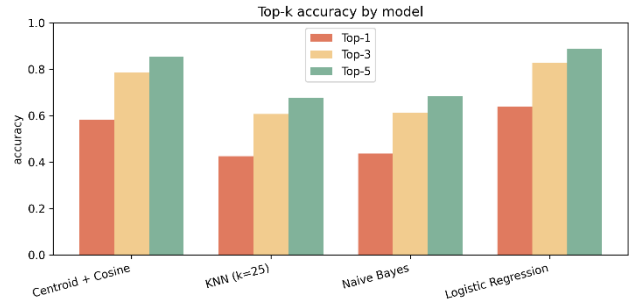


Fig. 2 Top-k accuracy of the four recommenders on the held-out test set.

B. Per-Model Discussion

The centroid model is the most interpretable: each journal is a single point in TF-IDF space, the inner product with the query gives the score, and one can read off the most influential terms. KNN delivers slightly weaker accuracy but has the additional advantage of returning, alongside the journal list, the actual articles that drove the recommendation, a useful debugging signal. Logistic regression is the strongest discriminative model when classes share vocabulary, while multinomial Naive Bayes is fastest to train but slightly weaker because of its strong conditional-independence assumption.

C. Topic-Cluster Discovery

Sweeping k in $\{4, 6, \dots, 24\}$ and scoring each clustering by silhouette on a 1,000-document subsample yielded $k^* = 14$ (Fig. 3). The resulting 14 clusters' top-12 terms map onto recognisable computer-science sub-areas: telecommunications and electronics (telecommunications, electrical, channel); semantic web and ontologies (web, semantic, ontology, services); information science and library science (library, information, business, systems); operations research and management science (operations research,

management, research); finite-element and numerical methods (method, mathematics, finite element, equations); hardware architecture and VLSI (hardware, architecture, performance); software engineering, HCI and education (computer science, software, engineering, systems); optimisation and evolutionary algorithms (optimization, algorithm, genetic, evolutionary, swarm); theoretical computer science (theory, mathematics, logic); cloud computing and security (cloud, computing, security, service); computational biology (biology, molecular, gene, protein); wireless sensor networks (sensor, wireless, networks, nodes, energy, routing); fuzzy decision making (fuzzy, decision, sets, linguistic, rough); and AI, neural learning and image processing (artificial intelligence, learning, image, data).

The cluster $\times$ journal cross-tab confirms that journals with the same declared scope concentrate inside the same cluster: the telecommunications cluster is dominated by IEEE TRANSACTIONS ON COMMUNICATIONS, IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS and JOURNAL OF COMMUNICATIONS TECHNOLOGY AND ELECTRONICS; the wireless sensor cluster by ACM TRANSACTIONS ON SENSOR NETWORKS, WIRELESS NETWORKS and AD HOC & SENSOR WIRELESS NETWORKS; the optimisation cluster by IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION, EVOLUTIONARY COMPUTATION and JOURNAL OF HEURISTICS; the theoretical CS cluster by INFORMATION AND COMPUTATION, DESIGNS CODES AND CRYPTOGRAPHY and SIAM JOURNAL ON COMPUTING; the AI cluster by NEURAL NETWORKS, COMPUTER SPEECH AND LANGUAGE and MEDICAL IMAGE ANALYSIS. This is the standard external-validation argument for a clustering result.

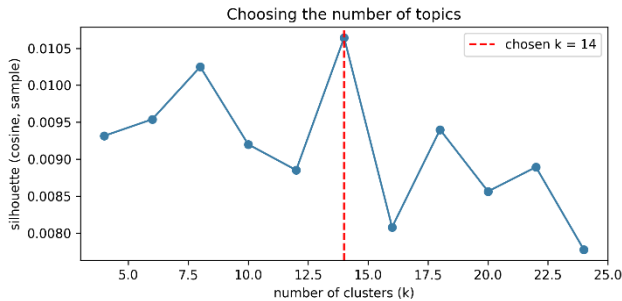


Fig. 3 Silhouette score (cosine distance, 1 000-document subsample) over candidate k . The maximum at $k = 14$ was used to fit the final clustering.

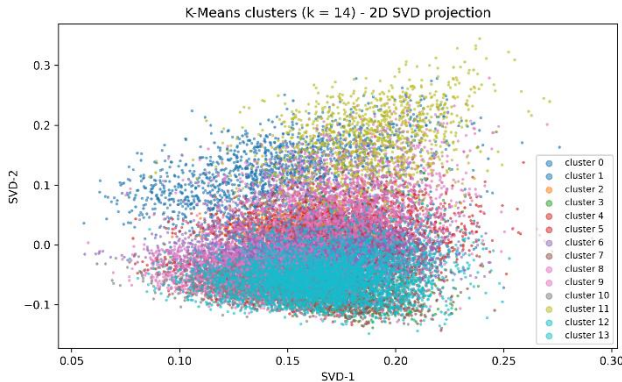


Fig. 4 Two-component truncated-SVD projection of the TF-IDF vectors, coloured by K-Means cluster ($k = 14$).

VI. CONCLUSION AND FUTURE WORK

We have built a complete journal-finder for the CompSciencePub corpus that meets the assignment's specification: it accepts a free-text abstract and returns a ranked top-5 list of computer-science journals. Four standard content-based recommenders were evaluated on a stratified 80/20 split of 23,034 abstracts across 428 journals; logistic regression gave the strongest combination of Top-5 accuracy (0.888) and MRR (0.737), with the centroid + cosine baseline a close and considerably cheaper second (0.855 / 0.689). A complementary unsupervised analysis with K-Means ($k^* = 14$, chosen by silhouette) exposed 14 coherent research-area clusters and confirmed the topical structure of the corpus.

Several directions remain open. First, replacing TF-IDF with SciBERT or SPECTER embeddings is likely to improve performance on out-of-corpus queries, at the cost of GPU inference. Second, incorporating journal-level metadata such as impact factor, scope statement and acceptance rate would let the system trade off topical fit against publication-pipeline considerations. Third, citation-graph features as in [8] can help disambiguate journals whose abstracts look similar. Finally, a learning-to-rank framework [9] would let us combine the centroid, KNN, NB and logistic-regression scores into a single calibrated ranker.

REFERENCES

- [1] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988.
- [2] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge University Press, 2008.
- [3] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proc. 5th Berkeley Symp. Math. Statist. Probab.*, vol. 1, 1967, pp. 281–297.
- [4] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent Dirichlet allocation," *J. Machine Learning Research*, vol. 3, pp. 993–1022, 2003.
- [5] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," *J. Comput. Appl. Math.*, vol. 20, pp. 53–65, 1987.
- [6] M. J. Schuemie and J. A. Kors, "Jane: suggesting journals, finding experts," *Bioinformatics*, vol. 24, no. 5, pp. 727–728, 2008.
- [7] D. Wang et al., "A content-based recommender system for computer science publications," *Knowledge-Based Systems*, vol. 157, pp. 1–9, 2018.
- [8] N. Kang, M. A. Doornenbal, and R. J. Schijvenaars, "Elsevier Journal Finder: Recommending journals for your paper," in *RecSys*, 2015, pp. 261–264.
- [9] T.-Y. Liu, "Learning to rank for information retrieval," *Foundations and Trends in IR*, vol. 3, no. 3, pp. 225–331, 2009.
- [10] Elsevier, "Journal Finder," <https://journalfinder.elsevier.com/>.
- [11] Springer Nature, "Journal Suggester," <https://journalsuggester.springer.com/>.
- [12] Wiley, "Journal Finder," <https://journalfinder.wiley.com/>.